

Abstract A

Target question: **Do autistic people perform better or worse than allistic people on statistical learning tasks?**

Impairments in statistical learning might be a common deficit among individuals with Specific Language Impairment (SLI) and Autism Spectrum Disorder (ASD). Using metaanalysis, we examined statistical learning in SLI (14 studies, 15 comparisons) and ASD (13 studies, 20 comparisons) to evaluate this hypothesis. Effect sizes were examined as a function of diagnosis across multiple statistical learning tasks (Serial Reaction Time, Contextual Cueing, Artificial Grammar Learning, Speech Stream, Observational Learning, and Probabilistic Classification). Individuals with SLI showed deficits in statistical learning relative to age-matched controls. In contrast, statistical learning was intact in individuals with ASD relative to controls. Effect sizes did not vary as a function of task modality or participant age. Our findings inform debates about overlapping socialcommunicative difficulties in children with SLI and ASD by suggesting distinct underlying mechanisms. In line with the procedural deficit hypothesis (Ullman and Pierpont, 2005), impaired statistical learning may account for phonological and syntactic difficulties associated with SLI. In contrast, impaired statistical learning fails to account for the social-pragmatic difficulties associated with ASD.

What did they do?	What did they find?	Critical evaluation

Abstract A is from: Obeid, R., Brooks, P. J., Powers, K. L., Gillespie-Lynch, K., & Lum, J. A. G. (2016). Statistical Learning in Specific Language Impairment and Autism Spectrum Disorder: A Meta-Analysis. *Frontiers in Psychology*, 7, 1–18. <https://doi.org/10.3389/fpsyg.2016.01245>

Abstract B

Target question: **Do autistic people perform better or worse than allistic people on statistical learning tasks?**

Background: Word learning is an important component of language development that influences child outcomes across multiple domains. Despite the importance of word knowledge, word-learning mechanisms are poorly understood in children with specific language impairment (SLI) and children with autism spectrum disorder (ASD). This study examined underlying mechanisms of word learning, specifically, statistical learning and fast-mapping, in school-aged children with typical and atypical development. **Methods:** Statistical learning was assessed through a word segmentation task and fast-mapping was examined in an object-label association task. We also examined children's ability to map meaning onto newly segmented words in a third task that combined exposure to an artificial language and a fast-mapping task. **Results:** Children with SLI had poorer performance on the word segmentation and fast-mapping tasks relative to the typically developing and ASD groups, who did not differ from one another. However, when children with SLI were exposed to an artificial language with phonemes used in the subsequent fast-mapping task, they successfully learned more words than in the isolated fast-mapping task. There was some evidence that word segmentation abilities are associated with word learning in school-aged children with typical development and ASD, but not SLI. Follow-up analyses also examined performance in children with ASD who did and did not have a language impairment. Children with ASD with language impairment evidenced intact statistical learning abilities, but subtle weaknesses in fast-mapping abilities. **Conclusions:** As the Procedural Deficit Hypothesis (PDH) predicts, children with SLI have impairments in statistical learning. However, children with SLI also have impairments in fast-mapping. Nonetheless, they are able to take advantage of additional phonological exposure to boost subsequent word-learning performance. In contrast to the PDH, children with ASD appear to have intact statistical learning, regardless of language status; however, fast-mapping abilities differ according to broader language skills.

What did they do?	What did they find?	Critical evaluation

Abstract B is from: Haebig, E., Saffran, J. R., & Ellis Weismer, S. (2017). Statistical word learning in children with autism spectrum disorder and specific language impairment. *Journal of Child Psychology and Psychiatry*, 58(11), 1251–1263. <https://doi.org/10.1111/jcpp.12734>